

Week 3: Combinatorics

Instructions: This handout is meant to go over some basic Combinatorics techniques and skills. The problems covered here are meant to be of the same level as the NMTC descriptive questions and above that. Problems marked at the end with (*) are a bit more tough (and fun!).

The problems covered here are meant to build your problem-solving skills, and as such, they are open-book. You are encouraged to use any textbook, video, or other online or offline resource as long as it doesn't give away the complete solution.

For each section, there are some worked examples. These examples are meant to show the applications of various techniques and how to approach more difficult problems. It is recommended that you try to attempt all the example problems before reading the solution.

1. Basics

This section is concerned with the basics of Combinatorics. We assume the reader is familiar with the concepts of permutations and combinations, the pigeonhole principle, stars-and-bars, and some induction.

We start with some simple examples.

Example 1.1 At least how many positive integers must be chosen from $1 - 100$ inclusive, to guarantee that there are two consecutive numbers?

Solution. Intuitively, 2 of the chosen numbers being consecutive means that there must be at least 1 even and at least 1 odd number. Thus you might think to proceed as follows:

The "worst" possible case occurs when the first 50 numbers you choose are all even, or all odd. If the first 50 numbers you choose are even, the 51st number must be odd, and vice versa. This gives an even number and an odd number, which must mean there are 2 consecutive numbers. Hence, we must choose at least 51 such numbers.

While the above solution is intuitive and straightforward, it is wrong. Why? It has to do with the fact that it very unclear as to what we mean by "the worst possible case". Indeed, it is almost always impossible to aptly define the term "worst possible" and so the solution itself becomes shaky and incomplete/incorrect. We instead proceed as follows.

Consider the sets $\{1, 2\}, \{3, 4\}, \dots, \{99, 100\}$. These are 50 sets, and each set contains a pair of consecutive integers. We know that 50 numbers do not work, because we could choose one number from each of the above sets, so no two numbers are consecutive. However, consider what happens if we choose the 51st number. Since there are only 50 sets, the Pigeonhole Principle tells us that there must be a set from which we chose 2 numbers. But this is the same as choosing 2 consecutive numbers, due to the way we constructed our sets. Hence, we must choose at least 51 numbers.

Example 1.2 Every point on a plane is colored either red or blue. prove that no matter how the plane is colored, there must exist 2 points at a distance of exactly 1 unit, such that the points have the same color.

Solution. Once again, we use the Pigeonhole Principle. But how? Notice that there are 2 colors, but infinitely many points on the plane. Thus, if we could choose 3 points, each at a distance of 1 unit from the other 2 points, we would be done. Thus, consider an equilateral triangle of side length 1. The triangle has 3 vertices, so 2 of these vertices must have the same color. Since they are at a distance of 1 unit, we are done.

Example 1.3 Given a unit square, show that if 5 points are placed anywhere inside, or on the boundary of this square, then there are two points which are at most a distance of $\frac{\sqrt{2}}{2}$ apart.

Solution. The fact that there are 5 points gives us a hint that we should try to construct 4 pigeonholes, such that if 2 points are in the same pigeonhole, they satisfy the constraint of the question.

After a bit of thought, it is easy to see that we can divide the square into 4 equal smaller squares. By the Pigeonhole Principle, 2 points must be in the same smaller square. These points can have a distance equal to at most

the diagonal of the smaller square, which is exactly equal to $\frac{\sqrt{2}}{2}$ **Note:** In the above solution, we assumed that each point was in only one of the 4 smaller squares. What happens if there is a point that is in more than one pigeonhole? How does the proof change?

Now, we attempt a slightly harder problem.

Example 1.4 Prove that from a set of ten distinct two-digit numbers (in the decimal system), it is possible to select two disjoint subsets whose members have the same sum.

Motivation. The question statement is quite vague. In order to get a sense for what it is saying, we try a simple case where our set of ten distinct numbers just consists of the numbers from 10-19. Thus we have $S = \{10, 11, \dots, 19\}$. We now want to choose 2 subsets such that their members have the same sum. In this case, we can simply choose the subsets $\{10, 19\}$ and $\{11, 18\}$. However, we may also choose the subsets $\{10, 18, 19\}$ and $\{14, 16, 17\}$, as both sets have a sum of 47. This is important because we may need to choose one subset to have say, 3 elements, while the other could have 5 elements, depending on what our original set was. It is important to see we could also have a subset with just 1 element, and in this case the sum would just be that element itself. We now must prove that this is possible in general. To do this, first note that what we want is for two subsets to have the same sum. Thus, it is natural to let the possible sums be our pigeonholes, and the possible subsets be our pigeons. There are 2^{10} subsets of any given set, and we know intuitively that the total possible sums is going to be less than this, but how do we prove it?

Solution. Rather than trying to find the total possible sums for a specific set of 10 elements, we consider it for any possible set of 10 elements. The smallest sum we could have is simply 10, in case $\{10\}$ is a subset of our original set. The largest possible sum we could have is if the original set is $\{90, 91, \dots, 99\}$, which gives a sum of 945. Thus, the total possible sums, i.e the total amount of pigeonholes is $945 - 10 + 1 = 936$.

Since we have 1024 subsets, and 936 possible sums, we must have at least 2 subsets, say A and B , with the same sum. Now we are done(almost). The question states that our 2 subsets must be disjoint. If A and B are disjoint, then we are done. But if they are not, then we must construct 2 disjoint sets with the same sum. But that's easy. We just remove all the common elements from A and B , and so the resulting two sets satisfy the question.

Note: We assumed above that the 2 sets we are left with at the end are both non empty. How can we be sure this is the case?

Problems

The problems are ordered roughly in order of difficulty though it may or may not be accurate. You can ask us for hints in the gc or ask for more problems if you aren't satisfied with the ones provided! The last few problems are a bit tough!

Problem 1.1 If a Martian has an infinite number of red, blue, yellow, and black socks in a drawer, what is the minimum number of socks that the Martian must pull out of the drawer to guarantee they have a pair?

Problem 1.2 At least how many integers must be chosen from 1-100, to guarantee that there are two integers with a difference of 5.

Problem 1.3 Every point in a plane is either red, green, or blue. Prove that there exists a rectangle in the plane such that all of its vertices are the same color.

Problem 1.4 Show that at any party there are two people who have the same number of friends at the party (assume that all friendships are mutual).

Problem 1.5 Show that among any $n + 2$ integers, either there are two whose difference is a multiple of $2n$, or there are two whose sum is a multiple of $2n$.

Problem 1.6 Let n be a positive integer. Show that there is a positive multiple of n whose digits (in the base 10 representation) are all 0's and 1's.

Problem 1.7 Choose any $(n + 1)$ -element subset from $\{1, 2, \dots, 2n\}$. Show that this subset must contain two integers that are relatively prime.

Problem 1.8 Let n be a positive integer. Show that if you have n integers, then either one of them is a multiple of n or a sum of several of them is a multiple of n .

Problem 1.9 Seven points are placed inside a regular hexagon with side length 1. Show that at least two points are at most one unit apart.

Problem 1.10 Inside a 1×1 square, 101 points are placed. Show that some three of them form a triangle with area no more than 0.01.

Problem 1.11 Consider an $n \times n$ chessboard and place $2n$ identical pieces at the centers of different squares. Show that no matter how this is done, 4 pieces among them must form the vertices of a parallelogram.

Problem 1.12 A chess player prepares for a tournament by playing some practice games over a period of eight weeks. She plays at least one game per day, but no more than 11 games per week. Show that there must be a period of consecutive days during which she plays exactly 23 games.

Problem 1.13 Prove that the points of a right triangle of side length 1 cannot be colored in four colors such that no two points at distance at least $2 - \sqrt{2}$ from each other receive the same color.

Problem 1.14 Color the plane in two colors. Prove that there will always exist an equilateral triangle with all its vertices of the same color.

Problem 1.15 Show that it is possible to color the plane in two colors in such a way that there cannot exist an equilateral triangle of side 1 with all vertices the same color.

Problem 1.16 Color the lattice points of the plane in two colors. Show that there exists a rectangle all of whose vertices are the same color. A lattice point is simply a point with integer coordinates, like $(3, 4)$, $(81, 2098)$ etc.

Problem 1.17 Five lattice points are chosen in the plane lattice. Prove that you can always choose two of these points such that the segment joining these points passes through another lattice point.

Problem 1.18 Inside a room of area 5, you place 9 rugs, each of area 1 and an arbitrary shape. Prove that there are two rugs which overlap by at least $1/9$.

Problem 1.19 People are seated around a circular table at a restaurant. The food is placed on a circular platform in the center of the table, and this circular platform can rotate (this is commonly found in Chinese restaurants that specialize in banquets). Each person ordered a different entrée, and it turns out that no one has the correct entrée in front of him. Show that it is possible to rotate the platform so that at least two people will have the correct entrée.

Problem 1.20 There are 51 senators in a senate. The senate needs to be divided into n committees such that each senator is on exactly one committee. Each senator hates exactly three other senators. (If senator A hates senator B, then senator B does 'not' necessarily hate senator A.) Find the smallest n such that it is always possible to arrange the committees so that no senator hates another senator on his or her committee.

Problem 1.21 Given a real number a , we define a sequence by $x_0 = 1$, $x_1 = x_2 = a$, and $x_{n+1} = 2x_n x_{n-1} - x_{n-2}$ for $n \geq 2$. Prove that if $x_n = 0$ for some n , then the sequence is periodic.

A periodic sequence is a sequence that repeats, such as:

1,3,4,5,6,4,4,1,3,4,5,6,4,4,...

Problem 1.22 Let $S = 1, 2, 3, \dots, 2n$. Show that if we choose $n + 1$ numbers from S , then there exist two numbers such that one is a multiple of the other. (*)

2. Coloring and Invariants/Monovariants

This section will go over the concept of invariants and monovariants, as well as coloring proofs. It is assumed that the reader is familiar with the following results and definitions.

1. An invariant is an object/quantity/property that does not change.
2. A monovariant is an object/quantity/property that changes in only one direction, i.e always increases or always decreases, or possibly remains the same

The following examples illustrate these concepts

Example 2.1 Is it possible to tile a chessboard with 2×1 dominoes, if we remove 2 of its diametrically opposite corners?

Solution. No. A standard chessboard has 32 black squares and 32 white squares. Removing the two opposite corners removes either 2 black squares, or 2 white squares. Without loss of generality, we may assume it removes 2 black squares. We thus have 30 black squares, and 32 white squares. A domino will always cover 1 black square and 1 white square. Thus, after placing 30 dominoes, we will run out of black squares, so there will be 2 white squares which remain untiled. **Note.** The question stated that we were tiling a chessboard, so the coloring was given. However, even if the question just stated that it was an 8×8 board, we could color the board ourselves and the proof would still hold.

Example 2.2 Can you tile a 10×10 board with 4×1 dominoes?

Solution. We color the columns with 4 different colors, A, B, C, D . This gives us 30 squares of color A , 30 of color B , 20 of color C , and 20 of color D . We analyze how these colors change when we place 4×1 tiles. If we place a 4×1 tile vertically, it covers 4 squares of the same color. If we place a 4×1 tile horizontally, it covers 1 square of each color.

This tells us that vertical tiles keep the amount of squares of a color $\pmod{4}$ fixed, while horizontal tiles reduce the amount of squares of *every* color $\pmod{4}$ by 1. Thus, if a tiling was possible, all colors would have 0 squares

remaining at the end. This implies that the number of squares of each color at the start would also be $0 \pmod 4$, which is impossible because A and B have 30 squares each, which isn't divisible by 4.

Example 2.3 Let n be an odd positive integer, and let the numbers $1, 2, \dots, 2n$ be written on a blackboard. At every step, we take two numbers a, b , erase them from the board, and write the number $|a - b|$ on the board instead. At one point, there will only be one number remaining on the board. Prove that this number is odd.

Solution. Trying small values of n will show you that if n is even, the last remaining number will be even, and if n is odd, the last remaining number will be odd. This suggests that there is some quantity that is odd at the start, and *remains* odd throughout, i.e there is an underlying invariant.

With very little information about the numbers, we consider the *sum* of all the numbers on the board at each step, and denote it by S_t . Thus, S_0 is the sum of all the numbers on the board at the start, S_1 is the sum of all the numbers on the board after 1 step, and so on.

Furthermore, when removing the numbers a, b we may assume without loss of generality, that $a \leq b$. We now look at how the quantity S_t changes. We remove the numbers a, b , and add the number $b - a$. Thus we get:

$$\begin{aligned} S_{t+1} &= S_t - a - b + (b - a) \implies \\ S_{t+1} &= S_t - 2a \end{aligned}$$

We see that the quantity S_t always changes by an even number, so its *parity* stays the same throughout. We know also that initially, $S_0 = 1 + 2 + \dots + 2n$, which equals $n(2n + 1)$. Since n and $2n + 1$ are both odd, their product is odd and so S_0 is odd. Thus, the sum of the numbers at the end will also be odd, but the sum of the numbers at the end is just that last remaining number, and we are done.

Example 2.4 At first, a room is empty. Each minute, either one person enters or two people leave. After exactly 3^{1999} minutes, could the room contain $3^{1000} + 2$ people?

Solution. We analyze what happens when there are n people in the room. After one minute, there will be either $n + 1$ or $n - 2$ people in the room. We

see that these values differ by exactly 3. We can argue further and say that at *any* time t , all possible values for the number of people in the room differ from each other by some multiple of 3.

In 3^{1999} minutes, one possible value for the amount of people in the room is 3^{1999} , if one person enters every minute. Since this is a multiple of 3, any other possible value for the amount of people in the room has to also be a multiple of 3. Since $3^{1000} + 2$ is not a multiple of 3, we see that this value is not achievable.

Example 2.5 If 127 people play in a singles tennis tournament, prove that at the end of tournament, the number of people who have played an odd number of games is even.

Solution. Your first instinct might be to assume that the tournament has some sort of format, for example, one in which every player plays another person exactly once, etc. But the problem does not specify this at all, and so any solutions with this assumption are incorrect. We instead think of what information we *do* have, and only one thing comes to mind. That is, every game is played by exactly 2 people, so if we count the total number of games, every game is counted twice.

We thus let g_i be the amount of games played by person i at the end of the tournament, and analyze the quantity

$$g_1 + g_2 + \cdots g_{127}$$

For starters, this sum is even. Additionally we are summing an odd number of terms. We look at what happens if the number of people who have played an odd number of games is odd. In that case, the sum of all those terms would be odd, while the sum of all the other terms would be even. But then our quantity above would be odd+even = odd, a contradiction. Hence we have proven what we set out to do.

Note. If you do not understand the above logic, consider what happens if we only have 3 terms, g_1, g_2, g_3 . Then, we assumed by contradiction that only an odd number of these terms is odd. So for example, we assume that only g_1 is odd, while g_2, g_3 are both even. But then our sum $g_1 + g_2 + g_3$ is odd, which would be a contradiction.

Example 2.6 Let $a_1, a_2, \dots, a_n$ be a permutation of the numbers $1, 2, \dots, n$. Prove that if n is odd, then the product

$$(a_1 - 1)(a_2 - 2) \cdots (a_n - n)$$

is an even number.

Solution. Similar to the previous two examples, we have very little information, so we consider the sum of all the terms in the product:

$$\begin{aligned} & (a_1 - 1) + (a_2 - 2) + \cdots + (a_n - n) \\ &= (a_1 + a_2 + \cdots + a_n) - (1 + 2 + \cdots + n) \\ &= (1 + 2 + \cdots + n) - (1 + 2 + \cdots + n) \\ &= 0 \end{aligned}$$

Thus, we took the sum of an odd number of integers, namely the numbers $(a_1 - 1), (a_2 - 2), \dots, (a_n - n)$ and got 0. This means that there must be at least one even number in these terms, due to similar logic as the above problem. Hence, the product of these terms contains an even number, and so the product is even.

Example 2.7 In an elimination-style tournament of a two-person game (for example, chess or judo), once you lose, you are out, and the tournament proceeds until only one person is left. Find a formula for the number of games that must be played in an elimination-style tournament starting with n contestants.

Solution. The number of people who are left in the tournament is an example of a monovariant, because after each game, the number of people decreases by 1. Thus, if we start with n people, the tournament must end in $n - 1$ games.

Example 2.8 23 friends want to play soccer. For this they choose a referee and the others split into two teams of 11 persons each. They want to do this so that the total weight of each team is the same. We know that they all have integer weights and that, regardless of who is the referee, it is possible to make the two teams. Prove that they all have the same weight.

Solution. Let $a_1, a_2, \dots, a_{23}$ be the weights of the people. We let

$$S = a_1 + a_2 + \dots + a_{23}$$

From the condition in the question, we know that if we remove any person, the rest of the people can be split into two teams, each with the same weight, i.e. $S - a_i = 2X$, which implies that S and a_i have the same parity, for all a_i . Thus, either all a_i are even, or all are odd.

We now proceed as follows. If all a_i are even, we then define a sequence b_i such that $b_i = a_i/2$. If all a_i are odd, we define a sequence b_i as $b_i = a_i - 1$. It is easy to check that this new sequence b_i also satisfies the conditions of the original problem, and that $b_i \geq 0$ for every value of i , so that this sequence b_i can be treated as a new set of weights. We also know that the sum of all these values of b_i is strictly smaller than the sum of all the values of a_i , *unless* all the values of a_i are zero.

If we keep repeating this step, we keep reducing the sum of the weights, which implies we must eventually reach a state where all of our new sets of weights are 0. This tells us that all the original weights were equal. (Why?)

Problems

The problems are ordered roughly in order of difficulty though it may or may not be accurate. You can ask us for hints in the gc or ask for more problems if you aren't satisfied with the ones provided! The last few problems are a bit tough!

Problem 2.1 Assume an 8×8 chess board with the usual coloring. You may repaint all squares (a) of a row or column (b) of a 2×2 square. The goal is to attain just one black square. Can you reach the goal?

Problem 2.2 There are three piles with n tokens each. In every step we are allowed to choose two piles, take one token from each of those two piles and add a token to the third pile. Using these moves, is it possible to end up having only one token?

Problem 2.3 Initially, we are given the sequence $1, 2, \dots, 100$. Every minute, we erase any two numbers u and v and replace them with the value

$$uv + u + v$$

Clearly, we will be left with just one number after 99 minutes. Does this number depend on the choices that we made?

Problem 2.4 There are a white, b black, and c red chips on a table. In one step, you may choose two chips of different colors and replace them by a chip of the third color. If just one chip will remain at the end, its color will not depend on the evolution of the game. When can this final state be reached?

Problem 2.5 Some 9 cells of a 10×10 grid are infected. A cell becomes infected if it has 2 or more neighbors infected. By neighbors, we mean the 4 adjacent squares, and not the diagonals. Is it possible for the whole grid to become infected?

Problem 2.6 There is a positive integer in each square of a rectangular table. In each move, you may double each number in a row or subtract 1 from each number of a column. Prove that you can reach a table of zeros by a sequence of these permitted moves.

Problem 2.7 Show that if every room in a house has an even number of doors, then the number of outside entrance doors must be even as well.

Problem 2.8 Consider nine lattice points in three-dimensional space. Show that there must be a lattice point on the interior of one of the line segments joining two of these points.

Problem 2.9 Three frogs are placed on three vertices of a square. Every minute, one frog leaps over another frog, in such a way that the “leapee” is at the midpoint of the line segment whose endpoints are the starting and ending position of the “leaper.” Will a frog ever occupy the vertex of the square that was originally unoccupied?

Problem 2.10 Start with the set $\{3, 4, 12\}$. You are then allowed to replace any two numbers a and b with the new pair $0.6a - 0.8b$ and $0.8a + 0.6b$. Can you transform the set into $\{4, 6, 12\}$?

Problem 2.11 The vertices of an n -gon are labeled by real numbers $x_1, \dots, x_n$. Let a, b, c, d be four successive labels. If $(a-d)(b-c) < 0$, then we may switch b with c . Decide if this switching operation can be performed infinitely often.

Problem 2.12 We strike the first digit of the number 7^{1996} , and then add it to the remaining number. This is repeated until a number with 10 digits remains. Prove that this number has two equal digits.

Problem 2.13 Each term in a sequence $1, 0, 1, 0, 1, 0, \dots$ starting with the seventh is the sum of the last 6 terms mod 10. Prove that the sequence $\dots, 0, 1, 0, 1, 0, 1, \dots$ never occurs.

Problem 2.14 Let $P_1, P_2, \dots, P_{1997}$ be distinct points in the plane. Connect the points with the line segments $P_1P_2, P_2P_3, P_3P_4, \dots, P_{1996}P_{1997}, P_{1997}P_1$. Can one draw a line that passes through the interior of every one of these segments?

Problem 2.15 Suppose you start with the numbers $0, 1, \sqrt{2}$. At each step you can choose a rational number q . Then, you replace one of the numbers in the list, a with the number

$$a + q(b - c)$$

where b, c are the other two numbers in the list. To illustrate, suppose we took $q = 5/4$, and $a = 0, b = 1, c = \sqrt{2}$. Then we would replace the number 0 with the number

$$0 + \frac{5}{4}(1 - \sqrt{2})$$

and so our list would become $\frac{5 - 5\sqrt{2}}{4}, 1, \sqrt{2}$.

Is it possible to get to the list $0, 2, \sqrt{2}$? (*)

Problem 2.16 We assign an integer to each vertex of a regular pentagon, so that the sum of all these integers is positive. If three consecutive vertices have assigned numbers x, y, z , respectively, and $y < 0$, we are allowed to change the numbers (x, y, z) to $(x + y, -y, z + y)$. This transformation is made as long as one of the numbers on the vertices of the pentagon is negative. Decide if this process always comes to an end. (***)